Database-System Assignment 02.

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BCS-3H

Question #1:

part (a):

```
Select DISTINCT c.customer.name
from Customer c
Where c.customer.id IN(
  SELECT p.customer-id
  from Purchase P
  JOIN Purchase-details pd ON p.purchase.id =
                                  pd.purchase.id
JOIN Books b ON pd.book-id = b.book.id
Where b.price > (select Avg(price) from Books));
```

part(b):

```
Select b1.title
from books b1
where b1.price >(
  Select Avg(b2.price)
  from Books b2
Where b2.publisher= b1.publisher);
```

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part(c):-

```
Select c.customer-name
from Customer c
Where c.customer IN(
Select DISTINCT p.customer-id
from purchase p
JOIN purchase-DETAILS pd ON pd.purchase-id = p.purchase-id
JOIN Books b ON pd.book-id = b.book-id)
AND c.customer-id NOT IN(
Select p.customer-id
from purchase p
JOIN Purchase-Details pd ON p.purchase-id = pd.purchase-id
JOIN Books b ON pd.book-id = b.book-id
Where b.publisher = 'Pearson');
```

part(d):-

```
Select b.title
from Books b
Where b.book-id NOT IN(
Select DISTINCT pd.book-id
from Purchase-Details pd);
```

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part(E).

```
Select a.author_name
  from Author a
  Where a.author_id IN(
    Select ba.author_id
      from Book_Authors ba.
   JOIN Books b ON ba.book_id = b.book_id
  Where b.price > (Select Avg(price) from Books));
```

part(F).

```
Select customer_id, total_quantity
from(
   Select p.customer_id,
        (Select sum(pd.quantity)
        from purchase_Details pd
     Where pd.purchase_id = p.purchase_id) as
total_quantity
   from purchase p) as customer_purchase
 GROUP BY customer_id, total_quantity
 HAVING sum(total_quantity) > 10;
```

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Question #2:-

part(A):-

```
Select ps.passenger_name, b.bus_name, r.source_city,
                                   r.destination_city
from Passenger ps
JOIN Ticket t ON ps.passenger_id = t.passenger_id
JOIN Buses b ON t.bus_id = b.bus_id
JOIN Routes r ON t.route_id = r.route_id;
```

part(B):-

```
Select ps.passenger_name, t.travel_date, t.fare
from Passenger ps
JOIN Tickets t ON ps.passenger_id = t.passenger_id;
```

part(C):-

```
Select b.bus_name, Count(t.ticket_id) as total_
from Buses b                                passenger
left Join Tickets t ON b.bus_id = t.bus_id
Group By b.bus_id, b.bus_name;
```

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part(D).

```
Select r.source_city, r.destination_city, ps.passenger_name
from Routes r
JOIN Tickets t ON r.route_id = t.route_id
JOIN Passengers ps ON t.passenger_id = ps.passenger_id;
```

part(E).

```
CREATE VIEW PassengerTravelSummary as
Select ps.passenger_name, b.bus_name, r.source_city,
       r.destination_city, t.travel_date
from Passengers ps
JOIN Tickets t ON ps.passenger_id = t.passenger_id
JOIN Buses b ON t.bus_id = b.bus_id
JOIN Routes r ON t.route_id = r.route_id;
```

part(F).

```
(i) ALTER TABLE Passengers add contact_number
    VARCHAR(15);

(ii) ALTER TABLE Tickets modify fare DECIMAL(10,2);

(iii) ALTER TABLE Buses DROP Column Capacity;

(iv) Drop TABLE Routes;
```

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Question #3:-

part(a):- Entities & their attributes:-

1. Workout-programs (Entity)

attributes:-

program-id (PK)
program-name
duration
difficulty-level.

2. Trainers (Entity)

attributes:

trainer-id (PK)
trainer-name
specialization
Experience.

3. fitness-center (Entity)

attributes:

center-id (PK)
center-name
location
Co-ordinator-name.

part(b):- Relationships & Cardinalities:-

1. Trainers ↔ Workout-programs (many to many)
2. Workout-programs ↔ fitness-center (M:N)

B

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part(c):
Since Trainers ↔ Workout programs is M:N, we must create an 'association entity'

i) Trainer-Program-Assignment:
trainer-id (PK), (FK)
program-id (PK, FK)
assignment-date
i) This table breaks M:N into two 1:M relationship
Similarly
Program-center:
program-id (PK, FK)
center-id (PK, FK)

part(d):
↳ Manager/Co-ordinatorName is not a separate entity because it doesn't have independent existence it depends on entirely on a 'fitness-center'

Question #4.

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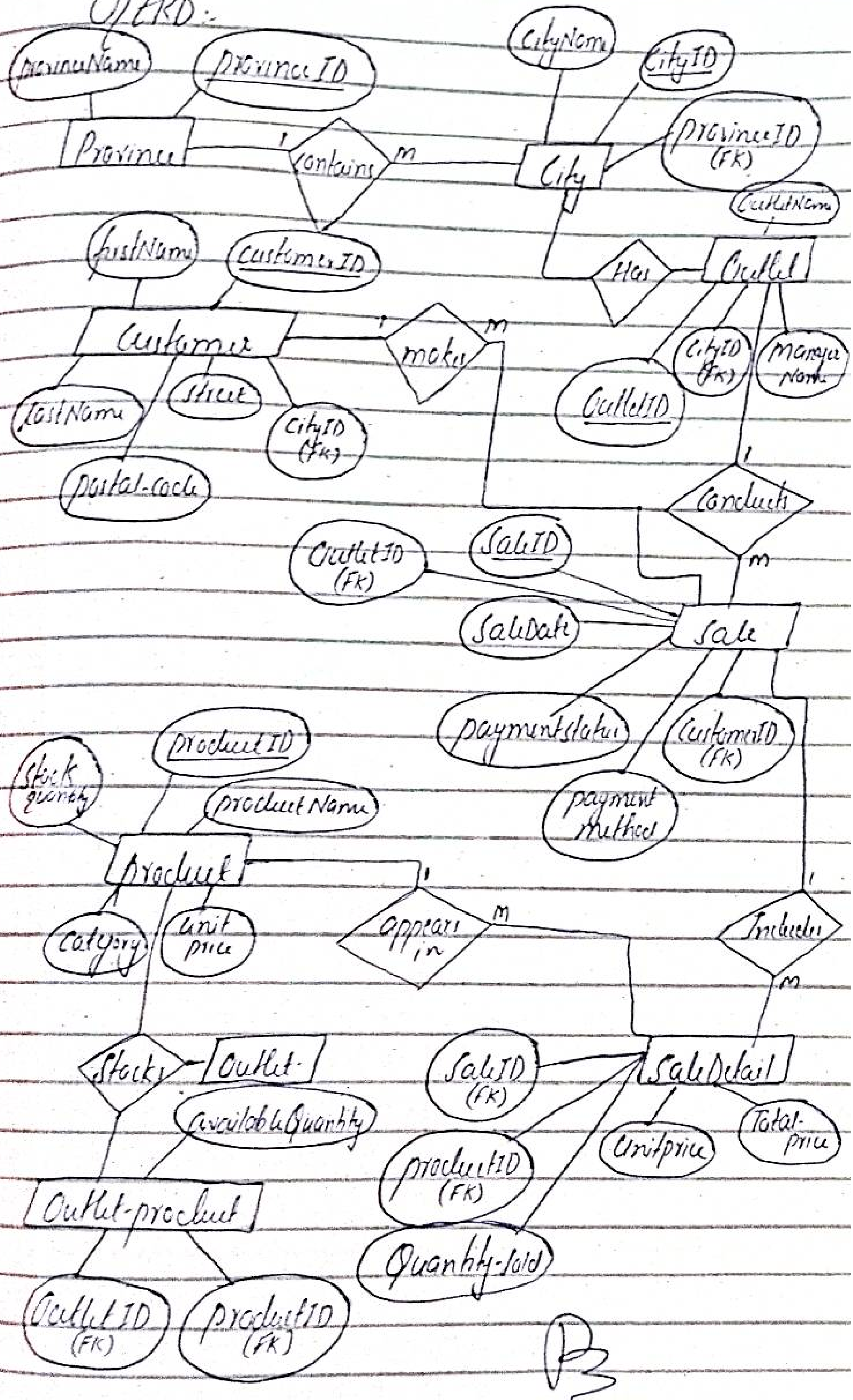
ERD:



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Question#5

1. Anomalies

Insertion-anomalies:-

- Can't add a new AI-model without associating it with an article.
- Can't add a new journalist without an article submission.

Updation-anomalies:-

- Updating agencies phone number requires multiple row-updates.
- Changing AI model version requires multiple row updates.

Deletion-anomalies:-

- Deleting an article might remove journalist or agency-information.
- Removing the last article from an AI-model deletes model information.

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~~2- functional Anomalies~~ :-

2- functional dependencies :-

- ArticleID → Title, Content, JournalistID.
- JournalistID → JournalistName.
- ModelID → ModelName.
- AgencyID → AgencyName, AgencyPhone.
- (ArticleID, ModelID, AgencyID, Verification) → Outcome, Confidence Score.

3- Normalization :-

1NF :- already in 1NF

2NF :- (i) Articles (Table / Entity)

attributes :-

- ArticleID (PK)
- Title
- Content
- JournalistID (FK)

(ii) Journalists (Entity)

attributes :-

- JournalistID (PK)
- Journalist Name.

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(iii) AI-models (Entity / Table)

attributes:-

ModelID (PK)

ModelName

(iv) Agencies (Table)

attributes:-

AgencyID (PK)

AgencyName

Agencyphone.

(v) Article-Analysis (Table):-

attributes:-

ArticleID (PK, FK)

ModelID (PK, FK)

AgencyID (PK, FK)

VerificationDate (PK)

Outcome

ConfidenceScore

c) Already in 3NF as there is no transitive dependencies.

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4. Data-Redundancy Problems

Problems:-

1. Agency-information:- If AgencyPhone was stored in Article Analysis (Updation-anomaly).

2. AI model information:- ModelName would be repeated for each analysis, causing updation-anomaly.

3. Journalist-information:- JournalistName would be repeated across articles.

c) These redundancies can cause:-
(i) Inconsistent data.
(ii) Wasted storage-space.
(iii) Slower query performance.
(iv) Complex Maintenance Operations.